AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-27 | Entry window 27 October 2026 (full day) | 30 records screened in scope

30 RECORDS IN SCOPE (169 EXCLUDED)

9 OF 10 SUBTOPIC BANDS POPULATED

5 EFFECT ESTIMATES WITH A CI

4 TIER-A RECORDS

1 RECORDS CARRIED ON METADATA ALONE

Signal of the day

PurpleAir dries its own sample: internal heating in the housing lowers detection-zone RH and removes aerosol water before the PMS5003 counts it.

Laboratory and field work (Santa Rosa, CA; Winston-Salem, NC, against a Teledyne T640x) measured air temperature in the particle detection zone of the Plantower PMS5003, standalone and inside a PurpleAir. The housing warms the zone more than a bare sensor; a wraparound USB heater pushed sample RH down to regulatory-monitor conditions and lowered reported $PM_{2.5}$ at high ambient RH. Even unheated, the PurpleAir read below a standalone PMS5003, consistent with partial removal of hygroscopic water. CFD resolved airflow, residence time and temperature inside the unit. **Where it is weak:** two field sites, no fitted growth factor or quantified bias reduction in the abstract; preprint. **Why it matters here:** humidity corrections are fitted on ambient RH, but the optical engine sees a warmed, drier sample whose RH depends on housing and airflow. Corrections fitted on bare-PMS data need not transfer to housed units.

What changed today

- **Trajectory source maps fail a negative control.** Riyadh, 10 years of METAR dust codes: PSCF/CWT run on *locally lifted* dust (no distant source needed) was *more* enriched over candidate source regions than suspended dust (permutation $p \leq 0.007$); wind matching erased most structure in both.
- **Paris plume optics change within hours.** Airborne plus ground: refractive index moved from $\sim$1.35–1.40 + 0.03–0.08i near source to $\sim$1.45–1.50 + 0.015–0.02i 2–6 h downwind; SSA and AAE rose. A single assumed refractive index in OPC mass conversion is wrong for one of the two.
- **Prenatal $PM_{2.5}$ and infant antibiotics, Scotland.** 498,123 births; observed vs 5 $\mu g\,m^{-3}$ counterfactual: controlled direct effect OR 1.12 (1.10–1.14), indirect via birthweight 0.99. The exposure IQR is only 6–8 $\mu g\,m^{-3}$; read as residual area confounding until replicated.
- **Coal heating drives Polish respiratory admissions.** 779k emergency COPD/asthma admissions: associations only in the heating season (adult asthma RR 1.100 per IQR $PM_{2.5}$); NO_2 null throughout.
- **ERF choice dominates burden.** BEST-COST, five countries: disease-specific ERFs gave 2–4$\times$ lower YLL than WHO all-cause ERFs (median ratio 0.42 for $PM_{2.5}$), with stable rankings.
- **Power shutoffs compound wildfire smoke.** California 2013–2019: severe PSPS plus +10 $\mu g\,m^{-3}$ wildfire $PM_{2.5}$ vs neither, respiratory encounters OR 1.16 (1.10–1.22), multiplicative interaction — the window in which mains-powered sensors and purifiers go dark.
- **Low-cost $PM_{2.5}$ calibration holds through winter; gases do not.** London, 10 months, two sites: linear T/RH calibration matched ML for $PM_{2.5}$, transferred at R^2 $\geq$0.56; gas calibrations transferred at R^2 $\leq$0.28.
- **Cross-day collisions.** 15 of this day's in-scope records were already issued in the 26 October issue and are logged here as duplicates.

Window continuity

The 22 October issue closed with `last_entry_date` at 22 October. A 30 October 00:21 UTC run harvested 23–28 October but stopped before screening; this issue was built on the 30 October 14:20 UTC (09:20 CDT) recovery run, which backfills 23–29 October day by day. This issue collects **27 October in full**, continuous with the 26 October issue. Consensus, Scholar Gateway and the trial watch for the backfill window are reported in the 29 October issue.

Corpus analytics

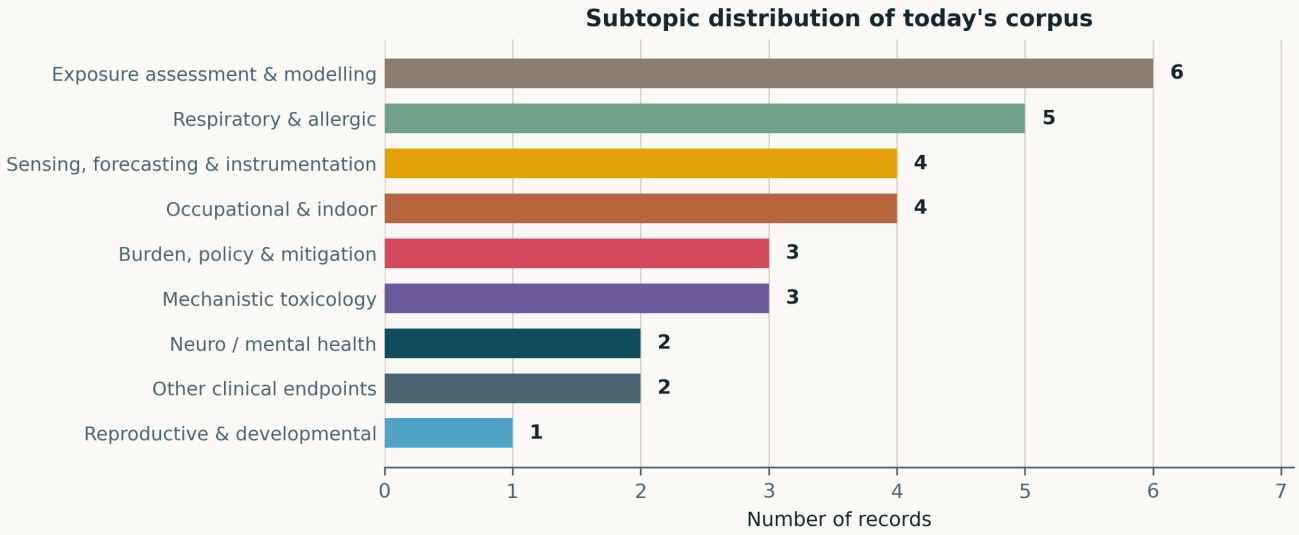


Fig. Subtopic assignment of the 30 in-scope records across nine bands. Exposure assessment leads with six (plume back-trajectories, trajectory negative control, Paris plume optics, Nepal on-road BC, Faisalabad PMF, Khulna); respiratory five; sensing and occupational/indoor four each; cardiovascular is the only empty band.

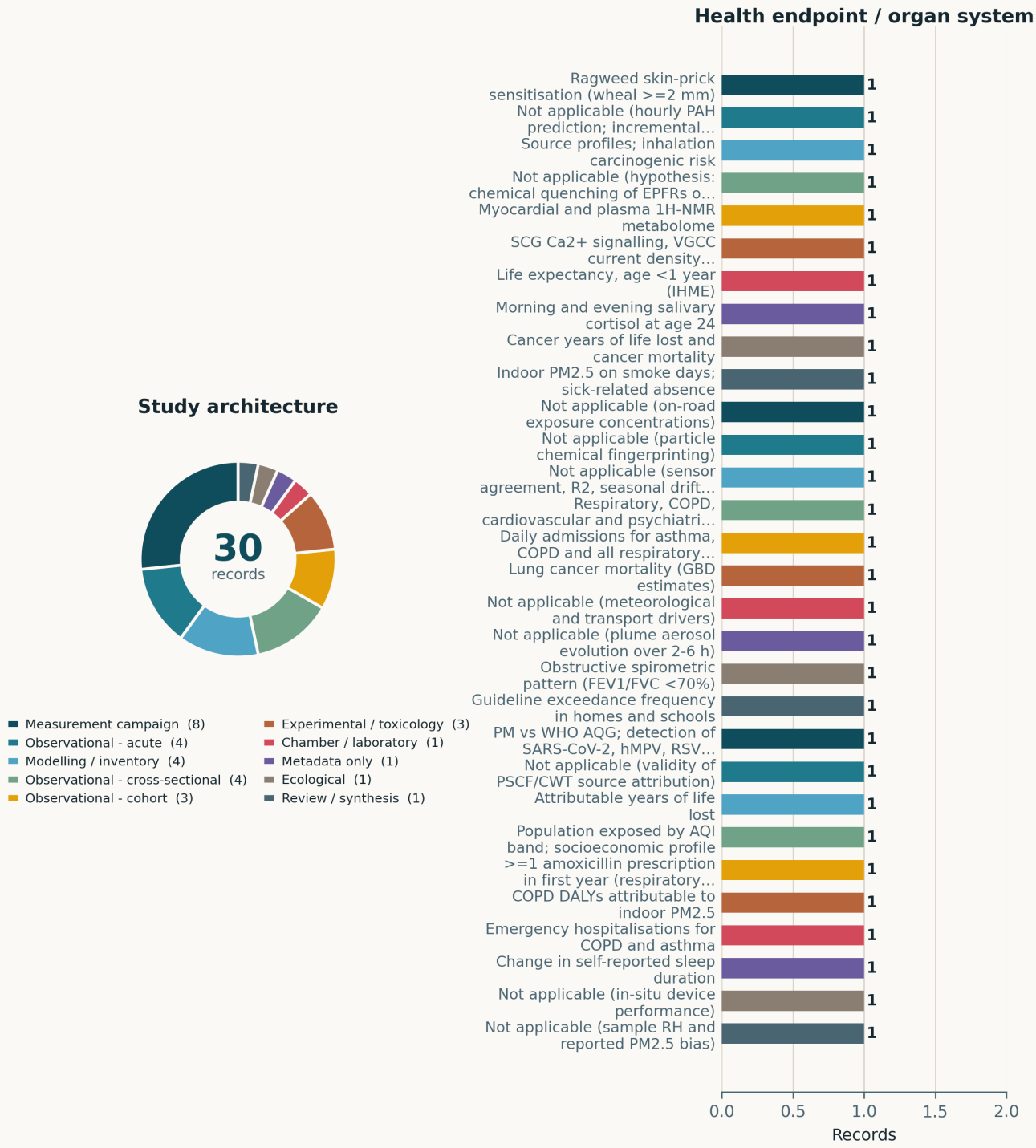


Fig. Left — architecture: time-series, case-crossover, cohorts and burden models alongside calibration, mobile, laboratory and monitoring analyses, two animal studies and one review; **metadata-only 1** (3 %). Nine records are *European Journal of Public Health* conference abstracts recovered by Crossref title match. Right — endpoints run from emergency admissions, spirometry and sensitisation to infant antibiotic use, absenteeism, cortisol and burden metrics.

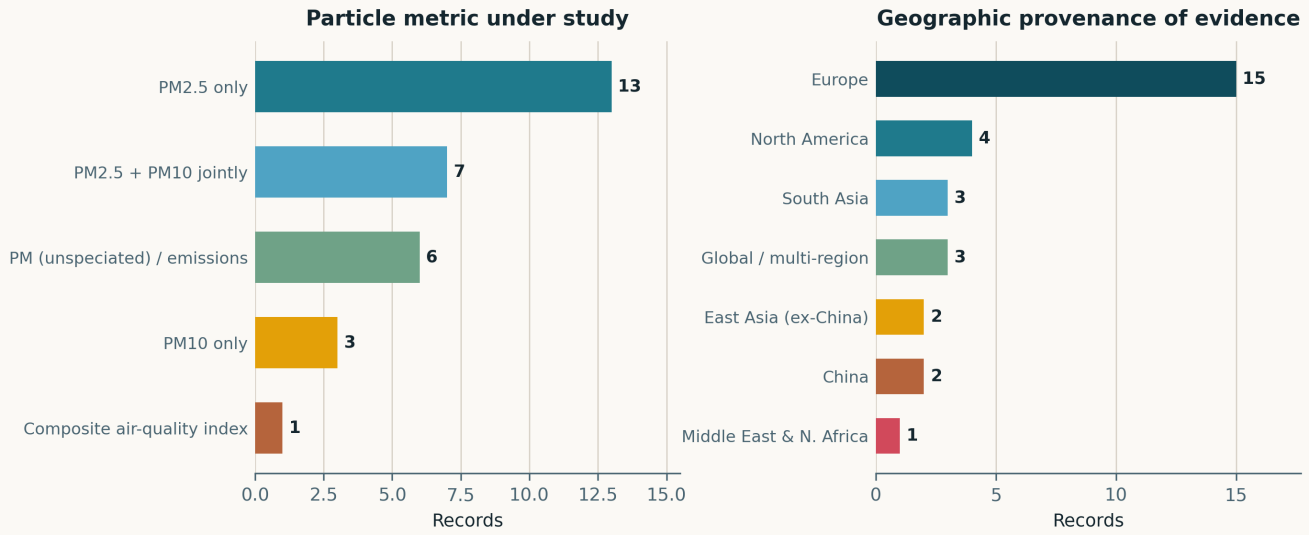


Fig. Left — particle metric: $PM_{2.5}$ dominates; PM_{10} (UK fire plume, Faisalabad, Croatia), eBC (Nepal), PM_1 (Paris) and dust codes (Riyadh) also appear. Right — geography: Europe carries the issue (UK, Poland, Spain, Greece, Ireland, Slovakia, France, Sweden, Croatia and a five-country burden study); the USA is second (California, Oklahoma, national infant life expectancy, PurpleAir heating), then South Asia and East Asia.

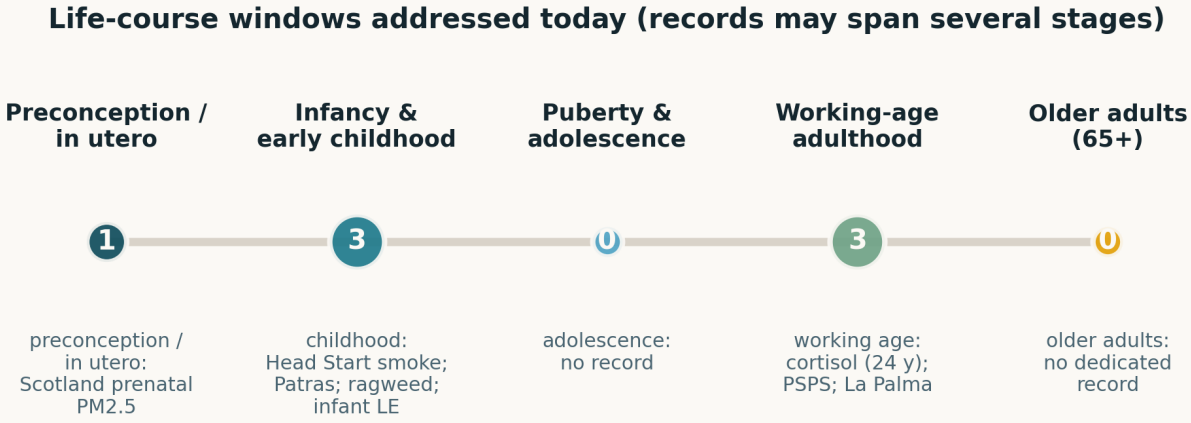


Fig. Life-course windows: one prenatal record (Scotland); three childhood records (Head Start smoke, Patras classrooms, Croatian ragweed sensitisation); three working-age records (cortisol at 24, PSPS, La Palma); no adolescent or older-adult record.

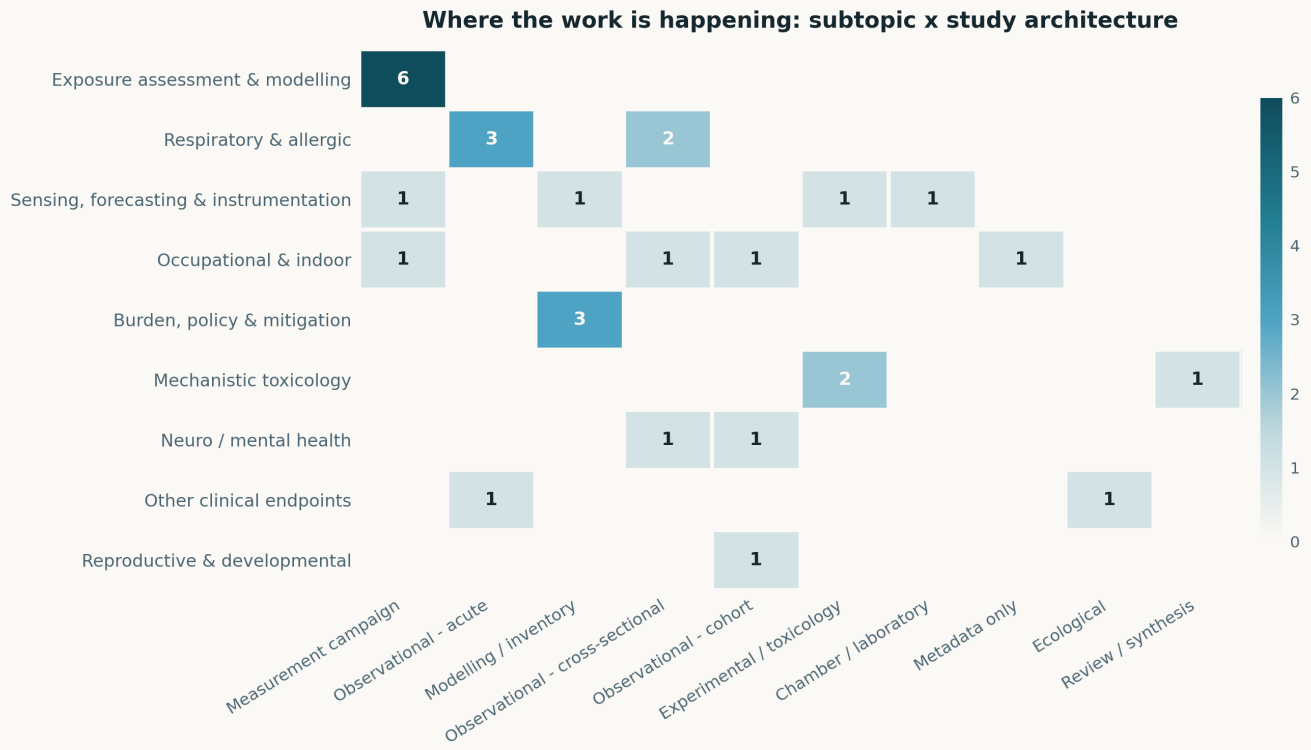


Fig. Subtopic × study architecture. Exposure assessment sits in campaign, monitoring and source-apportionment cells; sensing spans calibration comparison, laboratory instrumentation and ML; respiratory sits in time-series and case-crossover; mechanistic toxicology in two animal studies and one review.

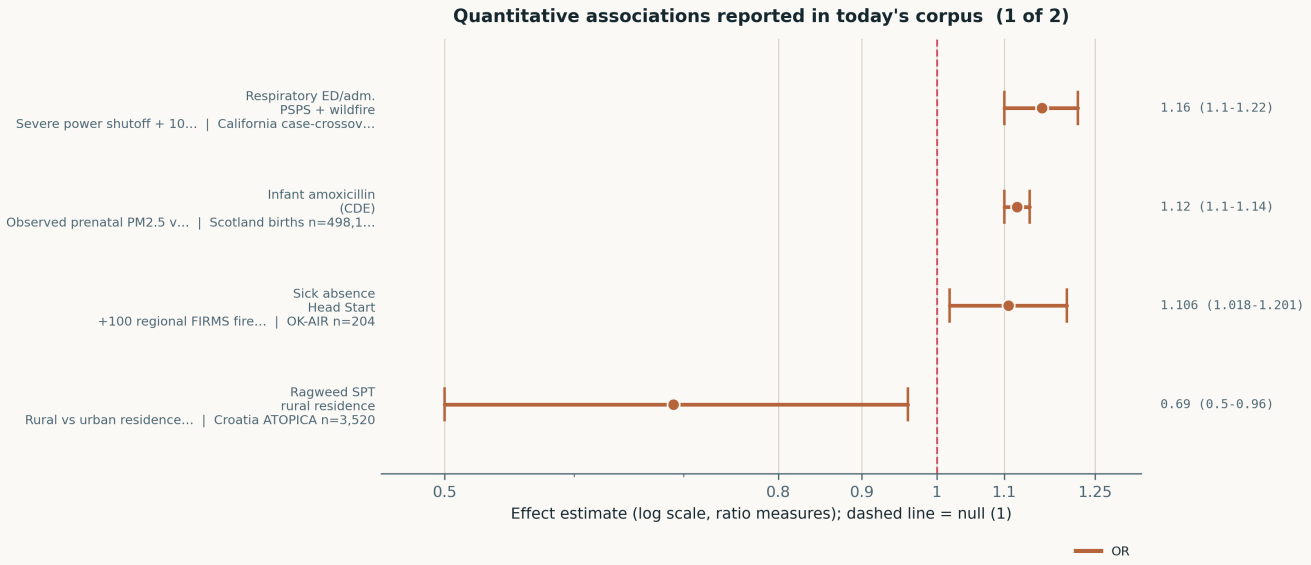


Fig. Ratio scale, four estimates from four studies: PSPS plus wildfire $PM_{2.5}$ and respiratory encounters (California); regional fire detections and sick absence (OK-AIR Head Start); observed prenatal $PM_{2.5}$ vs a 5 $\mu g\,m^{-3}$ counterfactual and infant amoxicillin (Scotland, controlled direct effect); rural residence and ragweed sensitisation (Croatia, area-level PM_{10}/pollen contrast). Exposure contrasts differ completely. Not poolable.

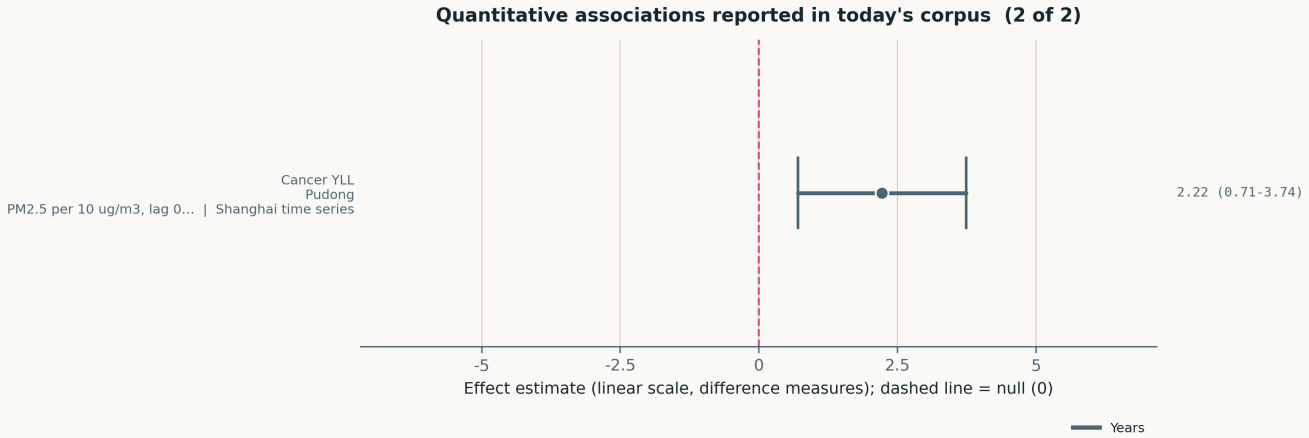


Fig. Difference scale: cancer years of life lost per $10\,\mu\mathrm{g\,m^{-3}}$ $\mathrm{PM}_{2.5}$ at lag 0–1, Pudong, Shanghai (2013–2021).

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 2 records

Li built environment sleep duration air pollution mediation Yinzhou cohort 2026 BMC Public Health

Association between built environment and sleep duration mediated by air pollution: a cohort study in China

Yinzhou cohort, 18,891 adults, baseline 2015-2017 and follow-up 2021-2023; quantile g-computation for a built-environment mixture and SEM for mediation. Per quartile of the mixture: +0.057 h/year sleep (0.039-0.075). Per IQR, PM_{10}, $PM_{2.5}$ and NO_2 were associated with -0.027, -0.048 and -0.071 h sleep. $PM_{2.5}$ mediated up to 21.5% of the green-space effect; NO_2 up to 45.3% of the walkability effect. Limitation: self-reported sleep, and SEM mediation needs no unmeasured mediator-outcome confounding - noise and light at night co-vary with traffic pollution and are only partly modelled. Relevance: low-moderate - mediation of this kind needs better spatial exposure, not more covariates.

doi: 10.1186/s12889-026-28717-y

Santoso air pollution noise greenness salivary cortisol BAMSE 2026 J Expo Sci Environ Epidemiol

Air pollution, road traffic noise, and greenness in relation to salivary cortisol in young adults

BAMSE birth cohort, n=1,148 at age 24; linear regression of salivary cortisol on prior-year modelled pollutants, noise and NDVI. No overall associations. In post hoc male subgroups, PM_{10} was associated with higher morning cortisol (beta 0.058, 0.003-0.112) as were NOx and noise, with greenness inverse; females showed opposite trends. Limitation: the only signals are exploratory sex-by-season subgroup contrasts after a null main analysis, with lower CIs at ~0; single-day cortisol is highly variable. Treat as null. Relevance: low - low-contrast Stockholm exposures modelled at the address.

doi: 10.1038/s41370-026-00995-z

Respiratory & allergic — 5 records

Poland air pollution COPD asthma emergency admissions DLNM heating season 2026 Eur J Public Health (conference abstract)

Air pollution and respiratory admissions in Poland: a nationwide DLNM analysis of COPD and asthma

National registry: 625,597 COPD and 153,563 asthma emergency admissions (32,893 paediatric), DLNM lag 0-10, stratified by heating season. Associations appeared only in the heating season: COPD RR per IQR $PM_{2.5}$ 1.044, PM_{10} 1.039, SO_2 1.055; lead 1.072, benzo[a]anthracene 1.057. Adult asthma stronger ($PM_{2.5}$ 1.100). In children, PM was not significant; arsenic (1.124) and ozone (1.115) were. NO_2 null throughout. No CIs in the abstract. Limitation: component and pollutant RRs come from single-pollutant models on collinear coal-combustion tracers, so 'lead' and 'PAH' effects cannot be separated from $PM_{2.5}$; conference abstract. Relevance: moderate - heating-season coal signatures are where speciation-capable nodes add value over mass.

doi: 10.1093/eurpub/ckag169.807

ISVOLCAN Tajogaite volcano La Palma lung function obstructive pattern 2026 Eur J Public Health (conference abstract)

Lung function after Tajogaite volcano eruption in La Palma: ISVOLCAN study

Random population sample of 604 adults (mean age 50; 62% women) with spirometry a median 424 days after the 85-day eruption. Obstruction prevalence 28.8%. Western (more exposed) residence: OR 1.64 (1.01-2.67); 1.74 (1.03-2.93) without prior respiratory disease. Intermediate SO_2 exposure OR 1.98 (1.18-3.34). Limitation: a fixed FEV1/FVC <70% cut-off over-calls obstruction in older adults, no pre-eruption baseline exists, and residence area is a coarse proxy that cannot separate SO_2 from volcanic PM; non-monotone SO_2 categories. Conference abstract. Relevance: moderate - volcanic events are a case for rapidly deployed PM and SO_2 sensor networks.

doi: 10.1093/eurpub/ckag169.069

Dublin air pollution fungal spores respiratory admissions time series 2026 Eur J Public Health (conference abstract)

Short-term effects of air pollution and aeroallergens on respiratory admissions

Dublin daily HIPE admissions with NO_2, $PM_{2.5}$, PM_1, fungal spores and meteorology; negative-binomial time series, lags 0-7. NO_2 was consistently associated with asthma, COPD and all respiratory admissions (~3-5% per unit, p <0.01); particulate effects were 'outcome-specific' and not quantified in the abstract; spores showed no independent short-term effect. Limitation: per-unit NO_2 effects of 3-5% are implausibly large per ppb or $\mu g\,m^{-3}$ and suggest an unstated scale; study years and PM results are missing; conference abstract. Relevance: low-moderate - Dublin's PM_1 series is unusual, and spore counts pair naturally with optical sensor coarse channels.

doi: 10.1093/eurpub/ckag169.1252

Wilner public safety power shutoff wildfire $PM_{2.5}$ healthcare California 2026 PLoS Med

Co-exposure to planned power outages and wildfire $PM_{2.5}$ and acute healthcare utilization in California: A time-stratified case-crossover study

Statewide California encounters for adults 20+, time-stratified case-crossover, temperature-adjusted. Severe PSPS events alone: same-day respiratory encounters OR 1.06 (1.02-1.11), stronger for COPD. Joint PSPS and +10 $\mu g\,m^{-3}$ wildfire $PM_{2.5}$ vs neither: OR 1.16 (1.10-1.22), with multiplicative interaction. Cardiovascular and psychiatric encounters null. Limitation: PSPS is circuit-level but assigned by ZIP, and outage plausibly changes the exposure itself (no power for filtration or HVAC, more time outdoors or relocation), so the interaction mixes effect modification with exposure modification. Relevance: high - outages disable mains-powered sensors and purifiers precisely during smoke; battery-backed nodes are needed to measure this window.

doi: 10.1371/journal.pmed.1005270

Lovric ragweed sensitisation PM_{10} pollen model comparison Croatia ATOPICA 2026 Children (Basel)

Outdoor Environmental Risk Factors for Paediatric Ragweed Sensitisation: A Formal Model Comparison

Re-analysis of 3,520 Croatian children from ATOPICA within EDIAQI; logistic models compared by AIC, adjusted for age and parental atopy. Sensitisation prevalence 10.8%. PM_{10} and categorical pollen area fit equally well (delta AIC 1.7) because both encode the same inland-coastal contrast; NO_2 added nothing. Transitional woodland-shrub aOR 0.93 (0.90-0.97) survived Bonferroni; rural residence aOR 0.69 (0.50-0.96). Limitation: the authors state it directly - area-level exposure cannot separate particulate pollution from pollen, so a PM contribution can be neither confirmed nor excluded. Relevance: moderate - an explicit case where only sub-regional, co-located PM and pollen measurement could break the collinearity.

doi: 10.3390/children13101366

Reproductive & developmental 1 record

Scotland prenatal $PM_{2.5}$ infant amoxicillin birthweight mediation 2026 Eur J Public Health (conference abstract)

Prenatal $PM_{2.5}$ and Infant Antibiotics: Birth-Weight Mediation via Causal Analysis

All Scottish singleton births 2009-2018, n=498,123, linked registries; g-computation mediation comparing observed $PM_{2.5}$ with a counterfactual of 5 $\mu g\,m^{-3}$. Controlled direct effect OR 1.12 (1.10-1.14); interventional direct OR 1.12 (1.11-1.14); indirect via birthweight 0.99 (0.99-1.01). Limitation: exposure IQR is only 6-8 $\mu g\,m^{-3}$, so a 12% excess over a ~2 $\mu g\,m^{-3}$ counterfactual shift is implausibly large and points to residual area-level confounding (deprivation, prescribing culture); amoxicillin dispensing is a weak infection proxy. Preliminary conference abstract. Relevance: low-moderate - fine-scale exposure contrast is too narrow for postcode annual means.

doi: 10.1093/eurpub/ckag169.955

Mechanistic toxicology 3 records

Do $PM_{2.5}$ superior cervical ganglion calcium channels satellite glia 2026 Int J Mol Sci

$PM_{2.5}$ Exposure Disrupts Sympathetic Neurotransmission by Impairing Voltage-Gated Ca2+ Channels and Remodeling Satellite Glia in the Superior Cervical Ganglion

Rats received intracerebroventricular $PM_{2.5}$; isolated superior cervical ganglia were also exposed directly. ICV $PM_{2.5}$ raised hypothalamic TNF-alpha and IL-6. Direct exposure cut medium norepinephrine without changing tissue content; depolarisation-evoked Ca2+ responses fell after either route, and patch clamp showed lower VGCC current density without Cav alpha1 mRNA change. GFAP and Kir4.1 rose (satellite glia). Limitation: intracerebroventricular injection bypasses the lung and does not model inhalation dose or translocation; particle source and dose relevance are not given in the abstract. Relevance: low.

doi: 10.3390/ijms27209184

Ganse diesel exhaust particles cardiac metabolomics glycolytic shift rats 2026 Biomolecules

Diesel Exhaust Particles Exposure Disrupts Cardiac Bioenergetics and Induces a Shift Towards Glycolytic Metabolism in Rats

Sprague-Dawley rats instilled with DEP or saline three times weekly for one or three weeks. At one week, 17 of 22 myocardial metabolites changed: succinate, acetate, formate, 3-hydroxybutyrate, ATP and ADP fell; glucose and lactate rose, consistent with reduced oxidative metabolism and a glycolytic shift, plus nucleotide degradation. Most changes resolved by three weeks; plasma changes were modest. Limitation: intratracheal bolus dosing of a single reference DEP does not reproduce inhaled dose rate, group sizes are not stated, and transient changes that resolve under continued exposure question chronic relevance. Relevance: low.

doi: 10.3390/biom16101501

Liu cerium dioxide scavenging environmentally persistent free radicals $PM_{2.5}$ filtration review 2026 Int J Mol Sci

Cerium Dioxide-Based Scavenging of Environmentally Persistent Free Radicals in Fine Particulate Matter: A Prospective Review

A hypothesis-stating review, explicitly not a validated result. EPFRs persist on $PM_{2.5}$ for days to months and drive hydroxyl-radical generation by Fenton-like cycling after deposition. The authors propose migrating CeO2 Ce3+/Ce4+ redox cycling, an established antioxidant mechanism, into ambient-temperature air filters to quench EPFRs on captured particles, and argue that high-temperature catalytic uses of CeO2 are not a precedent. They list the open questions: direct quenching vs ROS interception, humidity, airflow, contact time, and CeO2 fate. Limitation: no data; the only cited support is gas-phase ROS scavenging in cigarette smoke. Relevance: low - filter chemistry, not measurement.

doi: 10.3390/ijms27209232

Exposure assessment & modelling 6 records

UK Air Quality Cell PM_{10} plume back trajectory tyre fire Mexborough vulnerability 2026 Eur J Public Health (conference abstract)

Back trajectory modelled PM_{10} plumes: population vulnerability profile from an uncontrolled fire

The UK Air Quality Cell monitored PM_{10} during an uncontrolled tyre fire; 22 hourly ADMS-STAR back-trajectory plumes were spatially averaged and overlaid on Output Areas. 24,952 residents fell in 'Unhealthy for Sensitive Groups' or worse; 7,856 (31.5%) in 'Hazardous' or beyond. Exposed areas had above-median deprivation, prior ill health and lower education. The modelling suggested field monitoring under-captured peaks. Limitation: AQI bands from a 24 h index were applied to hourly plume concentrations, and exposure is residential, ignoring sheltering and evacuation; no health outcomes. Conference abstract. Relevance: high - rapid-deploy sensor arrays around industrial fires would close the peak-capture gap the authors identify.

doi: 10.1093/eurpub/ckag169.809

Aloraini negative control PSCF CWT trajectory dust attribution Riyadh 2026 Atmosphere

Transport Pathways Versus Dust Sources: A Negative-Control Test of Trajectory-Statistical Attribution in Riyadh, Saudi Arabia

Ten years of Riyadh METARs separated locally lifted dust (BLDU/BLSA, 179 days) from suspended dust (DU/SA, 345). ERA5 models predicted local lifting better (held-out AUC 0.914 vs 0.765). PSCF and CWT fields from 14,612 HYSPLIT trajectories were then run on the local class as a negative control: it was more enriched over a priori source regions than the suspension class (permutation p <=0.007), and wind matching erased most structure in both. Dust days also fell 85.9% in 2023-2024 without model explanation. Limitation: METAR codes are visibility-based, not PM. Relevance: high - trajectory 'source maps' from single receptors, common in LCS papers, mostly reflect transport climatology.

doi: 10.3390/atmos17111055

Yu Paris urban plume aerosol composition optical properties airborne ground 2026 Atmos Chem Phys

Urban-to-downwind variations in aerosol composition and optical properties in the Paris urban plume from coordinated airborne and ground-based observations

Coordinated ground and airborne measurements in Paris and its downwind plume, ~2-6 h transport. The plume modestly raised 80-200 nm particle number and moderately enhanced PM_1; organics dominated PM_1 throughout, with more oxidised organics downwind. Optical properties changed sharply: higher SSA at 450 and 630 nm downwind; refractive index moved from ~1.35-1.40 + 0.03-0.08i near source to ~1.45-1.50 + 0.015-0.02i downwind; AAE rose. Limitation: limited flights under specific meteorology; plume age is estimated, not tracked. Relevance: high - optical LCS use a single assumed refractive index; a 2-6 h change of this size shifts OPC mass conversion between urban and downwind nodes.

doi: 10.5194/acp-26-15055-2026

Upaul Khulna $PM_{2.5}$ meteorology HYSPLIT random forest STL 2026 Research Square (preprint)

Meteorological and HYSPLIT-Derived Transport Influences on $PM_{2.5}$ Exposure in a Khulna City, Bangladesh

One year (2024) of hourly $PM_{2.5}$ at a single Khulna station, analysed with MLR, Random Forest, STL and K-means on HYSPLIT trajectories. Winter mean ~121 $\mu g\,m^{-3}$ vs ~15 in the monsoon. Temperature and pressure were top RF predictors (OOB R2 0.64); temperature and RH were negative in MLR. Winter air arrived on short north-westerly continental paths, monsoon air on long southerly maritime paths; STL residuals flagged pre-monsoon spikes (~138 $\mu g\,m^{-3}$). Limitation: one station, one year, and reanalysis meteorology at ~50 km; season is collinear with every predictor, and the 'variance explained' shares are not additive. Preprint. Relevance: low-moderate - a coastal South Asian city with almost no monitoring, where LCS would add the spatial dimension.

doi: 10.21203/rs.3.rs-11078552/v1

Dhital on-road black carbon PM exposure Nepal route travel mode 2026 J Air Waste Manag Assoc

On-road exposure to black carbon and particulate matter in Nepal: Effects of route attributes and travel modes

Mobile monitoring across Nepali road types, gradients and travel modes. Mean on-road eBC 45.7 +/- 32.5 $\mu g\,m^{-3}$, ~11x background; eBC made up 51% of on-road $PM_{2.5}$ and 75% of PM_1. Concentrations rose from residential streets to intercity highways. Upgraded orbital-highway segments had lower $PM_{2.5}$ and PM_{10} but unchanged eBC and PM_1, so paving cut resuspended dust rather than exhaust. High-gradient (4.5%) segments: median eBC 86.7 vs 41.5 $\mu g\,m^{-3}$ on 0.6% segments. Motorcycles gave the highest exposure, then non-AC and AC buses. Limitation: no n of runs or season control is reported, and the old-vs-upgraded comparison is not a before-after design. Relevance: high - shows optical $PM_{2.5}$ nodes miss the exhaust-dominated BC contrast that drives on-road exposure.

doi: 10.1080/10962247.2026.2754932

Alvi PMF trace elements PM_{10} Faisalabad health risk 2026 J Hazard Mater

PMF Resolved Source Identification & Health Risk Assessment of Trace-elements in Airborne Particulate Matter (PM_{10}) from South-Asian Megacity

Filter PM_{10} in Faisalabad analysed by ICP-OES: mean 517 +/- 206 $\mu g\,m^{-3}$, ~10x WHO, higher in summer than autumn or winter. Element order Fe > Al > Si > Zn > Cu > Pb > Cd. PMF resolved seven sources: metal works, vehicle exhaust and non-exhaust, industrial fossil fuel, municipal waste incineration and industrial mixed dust. Cumulative carcinogenic risk 1.22e-3 (adults) and 5.05e-4 (children). Limitation: ICP-OES detection limits for As, Cr and Cd are marginal at these loadings, PMF with ~20 elements and no carbon or ions cannot separate combustion sources robustly, and n of samples is not given. Relevance: low-moderate - coarse, crustal-dominated PM_{10} at these levels saturates optical LCS assumptions.

doi: 10.1016/j.jhazmat.2026.144032

Sensing, forecasting & instrumentation 4 records

Ouimette PMS5003 PurpleAir internal heating hygroscopic mass reduction 2026 Preprints.org (preprint)

Air Flow, Heat Transfer, and Hygroscopic Aerosol Mass Reduction in the Plantower PMS5003 Sensor and PurpleAir Monitor

Lab and field work on temperature in the particle detection zone of the PMS5003, standalone and inside a PurpleAir. The PurpleAir housing warms the detection zone more than a bare PMS5003, lowering sample RH; a wraparound USB heater pushed sample RH down to regulatory-monitor conditions. In Santa Rosa and Winston-Salem (against a T640x), external heating lowered reported $PM_{2.5}$ at high ambient RH, and the unheated PurpleAir already read below a standalone PMS. CFD characterised airflow, residence time and temperature. Limitation: two sites, no quantified bias reduction or growth-factor fit in the abstract; preprint. Relevance: high - EPA-style RH corrections assume ambient RH, but the sensor sees a warmed, drier sample; corrections built on bare-PMS data may not transfer to housed units.

doi: 10.20944/preprints202610.2325.v1

Mercado-Velazquez low-cost sensors three-tier London calibration transferability 2026 Sensors (Basel)

How Accurate Are Low-Cost Air Quality Sensors? A 10-Month Study of Three-Tier Comparison with Three Calibration Methods and Cross-Site Transferability

Ten-month co-location of low-cost (MOS, optical), mid-cost electrochemical and reference instruments at two London sites. Calibration by manufacturer curve, T/RH linear regression and ML. Field-calibrated $PM_{2.5}$ explained over half of reference variance at both sites, and linear calibration matched ML. $PM_{2.5}$ calibrations trained April-October 2025 held through winter, whereas NO_2 and CO did not; transferred between sites, $PM_{2.5}$ reached R2 >=0.56 and gases R2 <=0.28. The mid-cost NO_2 and low-cost O_3 sensors failed under all methods. Limitation: two sites 1.6 km apart in one city share aerosol type and humidity regime, so 'transferability' is tested at its easiest. Relevance: high - supports simple T/RH regression for $PM_{2.5}$ nodes and mandatory co-location for gas channels.

doi: 10.3390/s26206441

Ai optical trapping Raman single carbonaceous aerosol smoke 2026 Sensors (Basel)

In-Situ Single-Particle Sensing of Carbonaceous Aerosols in Freshly Generated Smoke Using Optical Trapping-Raman Spectroscopy

Optical trapping-Raman spectroscopy captured smoke particles from nylon rope and plastic fibre ignited by the trapping laser itself, and analysed them airborne within seconds. Particles of several micrometres (~10 pg) gave full Raman fingerprints; band analysis separated two particle types, and spatially resolved spectra showed organic carbon and soot segregated within single particles. Limitation: freshly generated, supermicron, laser-ignited particles from two polymer fuels in an open chamber; ambient carbonaceous PM is mostly accumulation-mode, aged and far lighter, and the trapping laser may heat and alter what it measures. No throughput or detection-limit figures. Relevance: low-moderate - a reference technique for particle chemistry, not a field sensor.

doi: 10.3390/s26206527

Wang $PM_{2.5}$-bound PAH association rules machine learning forecasting Nagoya 2026 Environ Monit Assess

Characterizing statistical association patterns and local forecasting of $PM_{2.5}$-bound polycyclic aromatic hydrocarbon concentrations in Nagoya, Japan

Hourly $PM_{2.5}$-bound PAHs in Nagoya with association-rule mining and ML under time-blocked cross-validation. Low wind speed was the most consistent condition for PAH accumulation; BC and NOx peaked at rush hours. XGBoost and RF predicted high-molecular-weight PAHs best (R2 0.81-0.93, RMSE 0.02-0.11); SVR and MLP were poor. BaP dominated toxicity; lifetime cancer risks were below the USEPA negligible threshold. Limitation: a 10-day observation period, so the models learn one synoptic episode, and the authors themselves restrict the rules to site-specific description. Relevance: moderate - a BC-plus-meteorology proxy for PAHs is the kind of virtual sensor a network could run, if trained on far longer records.

doi: 10.1007/s10661-026-16083-6

Occupational & indoor — 4 records

Nahian Siegel in-situ testing in-duct air cleaners residential 2026 Build Environ

In-situ testing of in-duct air cleaners in residential buildings

METADATA ONLY. Title and authors only (Nahian, Siegel); no abstract retrievable at harvest from Crossref, Europe PMC or Semantic Scholar. The title describes in-situ testing of in-duct air cleaners (filters or electronic cleaners in forced-air systems) in occupied homes, presumably measuring particle removal under real operation. Country, number of homes, test protocol and removal metrics cannot be stated. Limitation: unverifiable beyond the title; in-situ effectiveness depends on HVAC runtime fraction, which lab single-pass efficiency ignores. Relevance: moderate - indoor LCS networks are the practical tool for verifying such devices in homes.

doi: 10.1016/j.buildenv.2026.115494

Patras primary school classroom PM airborne viruses ventilation 2026 Eur J Public Health (conference abstract)

Dual burden of PM and airborne viruses in primary schools: A pilot study in Patras, Greece

March 2026 sampling in 28 classrooms of seven primary schools. Mean $PM_{2.5}$ 37.8 and PM_{10} 109 $\mu g\,m^{-3}$; every classroom exceeded the WHO 24 h guideline, and at least one respiratory virus was detected in every room (SARS-CoV-2 most often). Frequent natural ventilation was associated with lower PM ($p < 0.033$, Cliff's delta >0.58) and less viral RNA. $PM_{2.5}/PM_{10}$ 0.35 points to resuspension. Limitation: short occupied-hours sampling compared against a 24 h guideline overstates exceedance, and one month in one city; instrument type unstated. Conference abstract. Relevance: moderate - coarse-dominated classroom PM is where optical LCS under-read most.

doi: 10.1093/eurpub/ckag169.1262

K-HEALTHinAIR multicountry indoor air homes schools low-cost sensors 2026 Eur J Public Health (conference abstract)

Multicountry evaluation of indoor air in homes and schools: results from K-HEALTHinAIR project

Low-cost sensors in homes and schools across four countries, 2025. Home $PM_{2.5}$ was elevated with smoking (93.9 $\mu g\,m^{-3}$), air conditioning (34.8), gas cooking (29.5) and Dutch district heating (19.4). School $PM_{2.5}$ was higher in Poland (22.0) than Austria (9.5). $PM_{2.5}$ exceedance reached 56% of days in Austrian wood-heated homes and 36% in Spanish smokers' homes; CO_2 exceedance was 41% in Austrian schools. Limitation: sensor calibration and RH correction are not described, so cross-country contrasts may partly be instrument and humidity artefacts; sampling of homes is unstated. Conference abstract. Relevance: high - a multi-country indoor LCS deployment whose QA is the open question.

doi: 10.1093/eurpub/ckag169.339

Fan wildfire smoke indoor $PM_{2.5}$ absenteeism Oklahoma Head Start OK-AIR 2026 J Expo Sci Environ Epidemiol

Wildfire smoke, indoor and outdoor $PM_{2.5}$, and sick-related absenteeism in Oklahoma Head Start early care and education centers

Cohort 1 of the OK-AIR cluster-randomised 2x2 factorial trial (HEPA, upper-room UVGI): 5 centres, 20 classrooms, 204 children aged 0-5, 2023-2024. On smoke-impacted days, the intervention period (pooled arms) was associated with 29.0% lower indoor $PM_{2.5}$ (22.8-34.7%) after adjustment for outdoor $PM_{2.5}$. Each +100 in the 7-day mean of regional FIRMS fire detections: sick-related absence OR 1.106 (1.018-1.201). Limitation: the period contrast pools all arms, so it is not a HEPA effect, and smoke days cluster in Feb-Mar, confounded by respiratory-virus season; fire counts are a regional proxy, not exposure. Relevance: high - classroom sensor plus EPA outdoor pairing is a deployable I/O design for childcare networks.

doi: 10.1038/s41370-026-00993-1

Burden, policy & mitigation — 3 records

K-HEALTHinAIR indoor $PM_{2.5}$ COPD attributable burden Spanish households 2026 Eur J Public Health (conference abstract)

Attributable Burden of COPD from Indoor $PM_{2.5}$ Exposure in Spanish Households, 2025

Comparative risk assessment combining measured household $PM_{2.5}$ (K-HEALTHinAIR), GBD 2023 Spanish rates and published exposure-response functions. Mean attributable COPD burden ~116 thousand DALYs; by household type, homes with smokers 327 thousand, air-conditioning for heating 183 thousand, no climate control 192 thousand, gas cooking 119 thousand. Limitation: applying ambient-derived ERFs to indoor $PM_{2.5}$ of different composition (tobacco, cooking) and scaling household-type estimates to national totals as if every home were that type; the subgroup figures are not additive and no uncertainty is given. Conference abstract. Relevance: moderate - burden estimates driven by low-cost indoor measurements.

doi: 10.1093/eurpub/ckag169.1053

BEST-COST all-cause vs disease-specific exposure-response functions burden $PM_{2.5}$ NO_2 2026 Eur J Public Health (conference abstract)

Effect of all-cause versus disease-specific exposure-response functions on air pollution burden

BEST-COST burden estimates (healthiar R package) in five countries, comparing WHO all-cause mortality ERFs with disease-specific ERFs. Disease-specific YLL were 2-4x lower (ratio 0.26-0.52; median 0.42 for $PM_{2.5}$, 0.31 for NO_2). Country rankings were stable (Spearman rho >=0.90), but absolute agreement was moderate for $PM_{2.5}$ (ICC 0.46) and poor for NO_2 (0.15). Limitation: five countries is too few for inference (Wilcoxon p=0.0625); the gap partly reflects causes of death excluded from the disease-specific set, which is a substantive question, not a method artefact. Conference abstract. Relevance: moderate - ERF choice, not exposure error, dominates burden uncertainty.

doi: 10.1093/eurpub/ckag169.810

Ruwali Lary US infant life expectancy environmental determinants machine learning 2026 Sensors (Basel)

Estimating Life Expectancy in the U.S.: Identifying Environmental Determinants and Disparities (2003 to 2019)

County-level IHME life expectancy for the under-1 age group, contiguous US 2003-2019, regressed on 33 ECMWF environmental variables with ML. Marked north-south disparity, southern counties lowest. With only five variables the model reached R2 0.81 and RMSE 1.06 years on an independent test set. Beyond PM, O_3 and CO, lower formaldehyde, sulphate and dust aerosol, more vegetation and lower wet-bulb temperature ranked as predictors of higher life expectancy. Limitation: no socioeconomic, race or healthcare covariates, so the south-north environmental gradient absorbs deprivation; feature importance is predictive, not causal, and IHME county estimates are themselves smoothed models. Relevance: moderate - a reanalysis-driven baseline that dense LCS data could sharpen.

doi: 10.3390/s26206478

Other clinical endpoints — 2 records

Slovakia $PM_{2.5}$ lung cancer mortality gender lag ecological 2026 Eur J Public Health (conference abstract)

$PM_{2.5}$ exposure and lung cancer mortality in Slovakia: gender differences and delayed effects

National time-series ecology, 2010-2023: population-weighted $PM_{2.5}$ fell from 24.4 to 15.2 $\mu g\,m^{-3}$ while lung-cancer mortality fell from 30.7 to 25.3 per 100,000 (AAPC -1.5%), driven by men (56.8 to 42.6; AAPC -2.2%); female rates were flat (12.6). Correlation with $PM_{2.5}$ r=0.95 overall and 0.97 in men, none in women; lags suggested emerging female associations. Limitation: 14 annual points of two monotonic trends correlate trivially; male smoking decline over the same period explains the male-only pattern, and the outcome is a GBD model output. Conference abstract. Relevance: low.

doi: 10.1093/eurpub/ckag169.1260

Zhong air pollution years of life lost cancer Pudong Shanghai time series 2026 PLoS One

The short-term impact of ambient air pollution on years of life lost from cancer in Pudong New Area, Shanghai: A time series study

Daily time series of cancer deaths and YLL in Pudong, 2013-2021, with DLNM (linear for YLL, quasi-Poisson for deaths). Per 10 $\mu g\,m^{-3}$ at lag 0-1: $PM_{2.5}$ +2.22 YLL (0.71-3.74), PM_{10} +1.04 (0.17-1.91); SO_2, NO_2 and O_3 also positive. Effects were marginally larger in men, younger and more educated people. Limitation: short-term harvesting of cancer deaths is mostly mortality displacement in terminally ill patients, and single-pollutant models leave the $PM_{2.5}$ estimate inseparable from co-pollutants; YLL per day is driven by a few young deaths, so the scale is unstable. Relevance: low - single-district monitor-average exposure.

doi: 10.1371/journal.pone.0360049

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Li built environment sleep duration air pollution mediation Yinzhou cohort	Association between built environment and sleep duration mediated by air pollution: a cohort study in China	*BMC Public Health*	Prospective cohort	Residential PM_{10}, $PM_{2.5}$, NO2 (modelled annual) as mediators	China (Yinzhou, Ningbo)
Santoso air pollution noise greenness salivary cortisol BAMSE	Air pollution, road traffic noise, and greenness in relation to salivary cortisol in young adults	*J Expo Sci Environ Epidemiol*	Cross-sectional (birth cohort)	Address-level $PM_{2.5}$, PM_{10}, black carbon, NOx (dispersion models), prior year	Sweden (Stockholm, BAMSE)
Reproductive & developmental					
Scotland prenatal $PM_{2.5}$ infant amoxicillin birthweight mediation	Prenatal $PM_{2.5}$ and Infant Antibiotics: Birth-Weight Mediation via Causal Analysis	*Eur J Public Health* (*conference abstract*)	Prospective cohort (registry linkage)	Annual postcode $PM_{2.5}$, pregnancy-month weighted (mean 7 $\mu g/m^3$)	United Kingdom (Scotland, 2009-2018)
Respiratory & allergic					
Dublin air pollution fungal spores respiratory admissions time series	Short-term effects of air pollution and aeroallergens on respiratory admissions	*Eur J Public Health* (*conference abstract*)	Time-series	Daily NO2, $PM_{2.5}$, PM1 (EPA Ireland); fungal spore counts	Ireland (Dublin)
ISVOLCAN Tajogaite volcano La Palma lung function obstructive pattern	Lung function after Tajogaite volcano eruption in La Palma: ISVOLCAN study	*Eur J Public Health* (*conference abstract*)	Cross-sectional (cohort baseline)	Residence area during the 2021 eruption; SO2 and PM exposure categories	Spain (La Palma, Canary Islands)
Lovric ragweed sensitisation PM_{10} pollen model comparison Croatia ATOPICA	Outdoor Environmental Risk Factors for Paediatric Ragweed Sensitisation: A Formal Model Comparison	*Children (Basel)*	School-based cross-sectional	Area-level PM_{10} and NO2; regional ragweed pollen; CORINE land cover (5 km)	Croatia (ATOPICA / EDIAQI)
Poland air pollution COPD asthma emergency admissions DLNM heating season	Air pollution and respiratory admissions in Poland: a nationwide DLNM analysis of COPD and asthma	*Eur J Public Health* (*conference abstract*)	Distributed-lag time-series	PM_{10}, $PM_{2.5}$, NO2, SO2, PAHs and metals in PM_{10} (national monitoring)	Poland (nationwide, 2012-2019)
Wilner public safety power shutoff wildfire $PM_{2.5}$ healthcare California	Co-exposure to planned power outages and wildfire $PM_{2.5}$ and acute healthcare utilization in California: A time-stratified case-crossover study	*PLoS Med*	Time-stratified case-crossover	ZIP-level daily wildfire $PM_{2.5}$ (per 10 $\mu g/m^3$) jointly with public safety power shutoffs	USA (California, 2013-2019)
Mechanistic toxicology					
Do $PM_{2.5}$ superior cervical ganglion calcium channels satellite glia	$PM_{2.5}$ Exposure Disrupts Sympathetic Neurotransmission by Impairing Voltage-Gated Ca2+ Channels and Remodeling Satellite Glia in the Superior Cervical Ganglion	*Int J Mol Sci*	Animal + in vitro	Collected $PM_{2.5}$, intracerebroventricular and ex vivo direct exposure	Korea (laboratory)
Ganse diesel exhaust particles cardiac metabolomics glycolytic shift rats	Diesel Exhaust Particles Exposure Disrupts Cardiac Bioenergetics and Induces a Shift Towards Glycolytic Metabolism in Rats	*Biomolecules*	Animal model	Diesel exhaust particles, intratracheal instillation 3x/week for 1 or 3 weeks	Spain (laboratory)
Liu cerium dioxide scavenging environmentally persistent free radicals $PM_{2.5}$ filtration review	Cerium Dioxide-Based Scavenging of Environmentally Persistent Free Radicals in Fine Particulate Matter: A Prospective Review	*Int J Mol Sci*	Narrative review	EPFRs on $PM_{2.5}$ surfaces; CeO2-functionalised filter media (proposed)	Not stated (review)
Occupational & indoor					
Fan wildfire smoke indoor $PM_{2.5}$ absenteeism Oklahoma Head Start OK-AIR	Wildfire smoke, indoor and outdoor $PM_{2.5}$, and sick-related absenteeism in Oklahoma Head Start early care and education centers	*J Expo Sci Environ Epidemiol*	Cohort (secondary trial analysis)	Continuous classroom $PM_{2.5}$ paired with hourly EPA outdoor $PM_{2.5}$; HMS smoke days; FIRMS fire detections	USA (Oklahoma)

First author	Focus	Journal	Design	Metric	Region
K-HEALTHinAIR multicountry indoor air homes schools low-cost sensors	Multicountry evaluation of indoor air in homes and schools: results from K-HEALTHinAIR project	*Eur J Public Health* (*conference abstract*)	Long-term indoor monitoring campaign	Low-cost sensor CO2, PM_{10}, $PM_{2.5}$ (annual 24 h means, 2025)	Europe (Spain, Austria, the Netherlands, Poland)
Nahian Siegel in-situ testing in-duct air cleaners residential	In-situ testing of in-duct air cleaners in residential buildings	*Build Environ*	Metadata only (no abstract)	Residential in-duct air cleaner performance (particle removal presumed)	Not stated (record metadata only)
Patras primary school classroom PM airborne viruses ventilation	Dual burden of PM and airborne viruses in primary schools: A pilot study in Patras, Greece	*Eur J Public Health* (*conference abstract*)	School-based cross-sectional	Classroom $PM_{2.5}/PM_{10}$ (triplicate) plus airborne virus RT-qPCR	Greece (Patras)
Sensing, forecasting & instrumentation					
Ai optical trapping Raman single carbonaceous aerosol smoke	In-Situ Single-Particle Sensing of Carbonaceous Aerosols in Freshly Generated Smoke Using Optical Trapping-Raman Spectroscopy	*Sensors* (*Basel*)	Laboratory instrument development	Single airborne smoke particles (several um, 10 pg), Raman spectra	Not stated (laboratory)
Mercado-Velazquez low-cost sensors three-tier London calibration transferability	How Accurate Are Low-Cost Air Quality Sensors? A 10-Month Study of Three-Tier Comparison with Three Calibration Methods and Cross-Site Transferability	*Sensors* (*Basel*)	Calibration model comparison	Low-cost optical $PM_{2.5}$ plus MOS/electrochemical NO2, CO, O3 vs reference	United Kingdom (London, two sites 1.6 km apart)
Ouimette PMS5003 PurpleAir internal heating hygroscopic mass reduction	Air Flow, Heat Transfer, and Hygroscopic Aerosol Mass Reduction in the Plantower PMS5003 Sensor and PurpleAir Monitor	*Preprints.org* (*preprint*)	Laboratory + field measurement	PMS5003 / PurpleAir $PM_{2.5}$ vs Teledyne T640x; detection-zone T and RH	USA (Santa Rosa, California; Winston-Salem, North Carolina)
Wang $PM_{2.5}$-bound PAH association rules machine learning forecasting Nagoya	Characterizing statistical association patterns and local forecasting of $PM_{2.5}$-bound polycyclic aromatic hydrocarbon concentrations in Nagoya, Japan	*Environ Monit Assess*	Machine-learning model	Hourly $PM_{2.5}$-bound PAHs, BC, NOx, oxidants, meteorology	Japan (Nagoya)
Exposure assessment & modelling					
Aloraini negative control PSCF CWT trajectory dust attribution Riyadh	Transport Pathways Versus Dust Sources: A Negative-Control Test of Trajectory-Statistical Attribution in Riyadh, Saudi Arabia	*Atmosphere*	Monitoring-record analysis	METAR present-weather dust codes (locally lifted vs suspension), 2016-2025	Saudi Arabia (Riyadh, Arabian Peninsula)
Alvi PMF trace elements PM_{10} Faisalabad health risk	PMF Resolved Source Identification & Health Risk Assessment of Trace-elements in Airborne Particulate Matter (PM_{10}) from South-Asian Megacity	*J Hazard Mater*	Source apportionment	Filter PM_{10} with ICP-OES trace elements; PMF	Pakistan (Faisalabad, Indo-Gangetic Plain)
Dhital on-road black carbon PM exposure Nepal route travel mode	On-road exposure to black carbon and particulate matter in Nepal: Effects of route attributes and travel modes	*J Air Waste Manag Assoc*	Mobile monitoring campaign	Equivalent BC (aethalometer), PM1, $PM_{2.5}$, PM_{10} (portable optical)	Nepal (Kathmandu Valley and intercity highways)
UK Air Quality Cell PM_{10} plume back trajectory tyre fire Mexborough vulnerability	Back trajectory modelled PM_{10} plumes: population vulnerability profile from an uncontrolled fire	*Eur J Public Health* (*conference abstract*)	Event-based monitoring	Incident PM_{10} monitoring + ADMS-STAR back-trajectory plume (22 hourly plumes)	United Kingdom (Mexborough, England)
Upaul Khulna $PM_{2.5}$ meteorology HYSPLIT random forest STL	Meteorological and HYSPLIT-Derived Transport Influences on $PM_{2.5}$ Exposure in a Khulna City, Bangladesh	*Research Square* (*preprint*)	Monitoring-record analysis	Hourly $PM_{2.5}$ from the CAMS-8 station (2024); NASA POWER meteorology; HYSPLIT	Bangladesh (Khulna)
Yu Paris urban plume aerosol composition optical properties airborne ground	Urban-to-downwind variations in aerosol composition and optical properties in the Paris urban plume from coordinated airborne and ground-based observations	*Atmos Chem Phys*	Aircraft measurement campaign	PM1 composition (AMS), particle number 80-200 nm, SSA, refractive index, AAE	France (Paris and downwind region)
Burden, policy & mitigation					
BEST-COST all-cause vs disease-specific exposure-response functions burden $PM_{2.5}$ NO2	Effect of all-cause versus disease-specific exposure-response functions on air pollution burden	*Eur J Public Health* (*conference abstract*)	Health impact model	EEA $PM_{2.5}$ and NO2 exposure; WHO-recommended ERFs	Europe (Belgium, Estonia, France, Norway, Portugal)
K-HEALTHinAIR indoor $PM_{2.5}$ COPD attributable burden Spanish households	Attributable Burden of COPD from Indoor $PM_{2.5}$ Exposure in Spanish Households, 2025	*Eur J Public Health* (*conference abstract*)	Burden estimation	Measured indoor $PM_{2.5}$ in Spanish homes (K-HEALTHinAIR)	Spain

First author	Focus	Journal	Design	Metric	Region
Ruwali Lary US infant life expectancy environmental determinants machine learning	Estimating Life Expectancy in the U.S.: Identifying Environmental Determinants and Disparities	*Sensors (Basel)*	Machine-learning model	33 ECMWF reanalysis environmental variables incl. PM, sulphate and dust aerosol, O3, CO, HCHO	USA (contiguous, county level)
Other clinical endpoints					
Slovakia $PM_{2.5}$ lung cancer mortality gender lag ecological	$PM_{2.5}$ exposure and lung cancer mortality in Slovakia: gender differences and delayed effects	*Eur J Public Health (conference abstract)*	Ecological	National population-weighted $PM_{2.5}$ (Indicator of Population Exposure), 2010-2023	Slovakia
Zhong air pollution years of life lost cancer Pudong Shanghai time series	The short-term impact of ambient air pollution on years of life lost from cancer in Pudong New Area, Shanghai: A time series study	*PLoS One*	Distributed-lag time-series	Daily $PM_{2.5}$, PM_{10}, SO2, NO2, O3 (monitoring network)	China (Shanghai, Pudong New Area, 2013-2021)

Provenance and method

Entry window **27 October 2026**, single day (Tuesday). Harvester legs (run 30 Oct 00:21 UTC): **PubMed** health **27**, sensing **6**; **Europe PMC 159** raw (25 AGRICOLA retro-index dropped); **Crossref by ISSN 49**; **arXiv** 150 kB, no in-window entry; **OpenAlex** HTTP 429. PubMed connector (EDAT 23–28 Oct, bucketed) 32 for this day. **199 screened**. DOIs and abstracts for 15 *Eur J Public Health* conference abstracts recovered by Crossref title match. **30 admitted**, **169 rejected** with logged reasons, of which 15 are cross-day duplicates of the 26 October issue and 3 within-window copies. **DOI gate** 0 fail / 0 warn. **1 carried on metadata alone**. Abstracts remain the copyright of their publishers.